

Governing Generative Landscape Design: A Formal Mathematical Framework for 4D Gaussian Splatting and Semantic Hypergraph Integration in Indigenous Heritage Reconstruction

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Abstract

Heritage reconstruction following the September 2023 Al-Haouz earthquake has exposed a structural limitation in contemporary spatial-AI tools: landscape models encode geometric data while abstracting away the governance relationships, oral histories, and collective land-tenure arrangements that give built heritage its social meaning. This paper introduces a layered mathematical framework that addresses this gap by unifying two complementary representations. The first layer is a four-dimensional Gaussian splatting model that captures the physical geometry of a High Atlas landscape across three temporal keyframes spanning the pre-earthquake, post-earthquake, and proposed-reconstruction states. The second layer is a typed governance hypergraph whose vertices enumerate structural spatial instances, human actors, oral-narrative records, Amazigh governance rules, and historical events, and whose hyperedges encode the relational and normative context of each element. We define a formal alignment map coupling these two layers, derive a joint embedding that fuses geometric and semantic information at the level of each structural instance, and formulate the reconstruction problem as constrained diffusion in Gaussian parameter space with a multi-term penalty objective whose governance loss is computed directly against the hypergraph constraint layer. We prove that conflict-preserving representation of contested ownership claims is necessary for governance-complete constraint satisfaction, show that the naïve weighted-sum constraint aggregation used in prior formulations loses relational dependency structure, and describe a community-validation active-learning loop by which field-collected annotations from Amazigh communities iteratively refine the hypergraph without collapsing contested claims. The framework is designed for on-site deployment in Morocco’s High Atlas and is accompanied by a fieldwork and oral-testimony collection protocol.

Keywords: 4D Gaussian splatting, semantic hypergraph, heritage reconstruction, indigenous governance, High Atlas, Amazigh, constrained generation, hypergraph neural networks, spatial AI, participatory design, Al-Haouz earthquake

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Notation

The following table collects the principal symbols used throughout the paper. All vectors are written in bold lower-case; all matrices in bold upper-case; all sets in calligraphic upper-case.

Symbol	Meaning
<i>Gaussian scene</i>	
$\mathcal{G}^{(t)}$	Gaussian splat scene at temporal keyframe t
g_k	A single Gaussian: $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{c}_k, \alpha_k)$
N	Total number of Gaussians in a scene
$\boldsymbol{\mu}_k \in \mathbb{R}^3$	Gaussian mean (world-space position)
$\boldsymbol{\Sigma}_k \in \mathbb{R}^{3 \times 3}$	Gaussian covariance (positive definite)
$\mathbf{c}_k$	Spherical-harmonic colour coefficients
$\alpha_k \in [0, 1]$	Gaussian opacity
$\mathbf{d}_k(t)$	Spatial deformation of Gaussian k at keyframe t
f_θ	Learned deformation network with parameters θ
$\mathcal{T}_{\text{kf}}$	Ordered keyframe set $\{t_0, t_1, t_2\}$
<i>Instance segmentation</i>	
$\mathcal{S} = \{s_1, \dots, s_n\}$	Set of structural instances
n	Number of structural instances
$\psi : [N] \rightarrow [n]$	Instance assignment map (Gaussian to structure)
$\mathcal{G}_{s_j}$	Gaussians assigned to instance s_j , i.e. $\{g_k : \psi(k) = j\}$
$\mathbf{g}_{s_j} \in \mathbb{R}^{d_g}$	Geometric feature vector of instance s_j
<i>Governance hypergraph</i>	
$\mathcal{H} = (V, \mathcal{E}, \phi_V, \phi_E)$	Typed governance hypergraph
V	Vertex set $\mathcal{S} \cup \mathcal{A} \cup \mathcal{N} \cup \mathcal{R} \cup \mathcal{V}$
$\mathcal{E} \subseteq \mathcal{P}(V)$	Hyperedge set (subsets of V)
$\phi_V : V \rightarrow \mathcal{T}_V$	Vertex type function
$\phi_E : \mathcal{E} \rightarrow \mathcal{T}_E$	Hyperedge type function
$\mathcal{A}$	Actor set (persons, lineages, institutions)
$\mathcal{N}$	Narrative set (oral testimonies, archival records)
$\mathcal{R}$	Rule set (Amazigh governance norms)
$\mathcal{V}$	Event set (historical episodes)
$\mathbf{B} \in \{0, 1\}^{ V \times \mathcal{E} }$	Hypergraph incidence matrix
$\mathbf{D}_V$	Diagonal degree matrix of vertices
$\mathbf{D}_E$	Diagonal degree matrix of hyperedges
$\mathbf{W}$	Diagonal hyperedge weight matrix
<i>Embedding and generation</i>	
$\mathbf{h}_v \in \mathbb{R}^{d_h}$	Hypergraph node embedding for vertex v
$\mathbf{z}_{s_j} \in \mathbb{R}^{d_z}$	Joint embedding of structural instance s_j
$\mathbf{Z}$	Matrix of all joint embeddings
$\mathcal{L}_{\text{recon}}$	Reconstruction fidelity loss
$\mathcal{L}_{\text{struct}}$	Structural/physical feasibility loss
$\mathcal{L}_{\text{gov}}$	Governance constraint violation loss
λ_1, λ_2	Non-negative penalty weights
$c_r(\mathcal{G}', \mathcal{H})$	Violation measure for rule $r \in \mathcal{R}$
$\Pi_{\mathcal{R}}$	Hard-constraint projection operator onto rule-feasible set

1 Introduction

1.1 The Al-Haouz earthquake and the reconstruction crisis

On 8 September 2023, a moment-magnitude 6.8 earthquake struck the Al-Haouz province of Morocco, with an epicentre approximately 70 kilometres south-west of Marrakech in the High Atlas mountains [1]. The event killed more than 2,900 people and severely damaged or destroyed tens of thousands of structures across a region whose built fabric is composed almost entirely of earthen construction. The Amazigh communities of the High Atlas, whose ancestors raised the kasbahs [قصبة], igherm [إيغرم] granaries, and terraced agricultural systems over many centuries, were disproportionately affected: rural villages in the Oukaïmeden, Azzaden, and N’Fis valleys suffered near-total structural collapse [1, 2].

The reconstruction challenge is therefore not primarily a structural engineering problem, although it includes one. It is, above all, a problem of reconstructing places in their social and cultural entirety: restoring not only the load-bearing walls but the thresholds at which collective water-sharing agreements were negotiated, the granary lofts whose allocation followed intergenerational inheritance protocols, and the sacred enclosures whose seasonal closure rules are still observed under the Agdal [أكڭال] governance system [3, 4]. No purely geometric description of a damaged building captures this information.

1.2 The shortcomings of current spatial planning tools

Contemporary digital reconstruction tools treat a landscape as a technical dataset. Photogrammetric pipelines and neural radiance fields [5] produce geometric reconstructions of impressive visual fidelity, but the output is a surface or a density field: a description of where matter is, not of what it means or who governs it. Geographic information systems (GIS) attach semantic labels to spatial objects, yet the label “building” carries no information about the lineage that built it, the communal well that constrains its setback, or the oral testimony of the elder who recalls its pre-earthquake form.

Three pathologies follow from this abstraction. First, planning decisions become centralised: because local knowledge is not encoded in the model, only agents with access to the model (typically external planners or state agencies) can reason about design options. Second, design proposals are evaluated against engineering codes alone, without recourse to the governance constraints that would determine whether a given layout is *permissible* within the community’s own normative framework. Third, the reconstruction erases contested or layered ownership claims by collapsing them into a single “current owner” attribute, producing designs that foreclose disputes that the community may wish to resolve on its own terms.

1.3 Spatial AI: a technological inflection point

The past three years have produced a rapid convergence of spatial AI capabilities that make a more ambitious approach possible. Three-dimensional Gaussian splatting [6] achieves real-time, photorealistic novel-view synthesis of complex scenes from collections of posed photographs and has been extended to dynamic scenes [7, 8]. Foundation segmentation models [9] enable open-vocabulary delineation of arbitrary structural instances from visual input. Graph neural networks [10, 11] and, more recently, hypergraph neural networks [12, 13] provide expressive relational reasoning over arbitrarily complex multi-party dependencies. Denoising diffusion models [14, 15] offer a principled framework for constrained generation.

None of these capabilities, taken individually, solves the reconstruction problem as stated. But their composition, under a shared formal framework that couples geometric and governance representations, opens a path toward a system that generates reconstruction proposals which are simultaneously geometrically plausible, physically sound, and governance-compliant.

1.4 The governance gap and this paper’s response

The formulation proposed in recent work on spatial AI for heritage reconstruction (summarised in Section 1.6) encodes governance metadata as a flat per-region tuple $(\mu_j, \alpha_j, \sigma_j)$, where μ_j is a permission flag, α_j is a single actor identifier, and σ_j is a scalar sensitivity score. This representation is inadequate for Amazigh governance for three reasons. First, Agdal land management involves collective institutions and multi-actor hyperedges: a single water-sharing rule binds a water channel, multiple terraced plots, and multiple household lineages simultaneously, a relationship that is not a binary edge between two nodes. Second, governance rules are conditional and temporal: certain enclosures are restricted only during the Agdal season, and restriction may be lifted by a collective decision of the village council. Third, ownership claims are frequently contested: multiple lineages may assert rights over the same plot, and the design system must represent this contestation without resolving it on behalf of the community.

This paper provides a formal response to all three inadequacies. We introduce a typed governance hypergraph whose hyperedges are first-class mathematical objects capable of representing multi-party, conditional, and temporal governance norms. We couple this hypergraph to a 4D Gaussian splatting model via an instance alignment map. We derive a joint embedding that carries both geometric and semantic information into the generative stage. And we prove (Proposition 4.18) that conflict-preserving hyperedge representation is not merely desirable but mathematically necessary for governance-complete constraint satisfaction.

1.5 Contributions

This paper makes the following contributions.

1. **4D Gaussian keyframe model (Section 3).** A formal specification of the spatial layer as a typed 4D Gaussian splat scene over three keyframes representing pre-earthquake, post-earthquake, and proposed-reconstruction states, together with a precise definition of the instance assignment map ψ .
2. **Typed governance hypergraph (Section 4).** A typed hypergraph $\mathcal{H} = (V, \mathcal{E}, \phi_V, \phi_E)$ with five vertex types and six hyperedge types, a formal treatment of temporal governance constraints, and a proof that conflict-preserving representation is necessary.
3. **Gaussian-to-hypergraph alignment (Section 5).** A joint embedding $\mathbf{z}_{s_j}$ that fuses the opacity-weighted geometric feature of each structural instance with its HGNN-derived semantic embedding.
4. **Constrained generation objective (Section 6).** A multi-penalty objective $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \lambda_1 \mathcal{L}_{\text{struct}} + \lambda_2 \mathcal{L}_{\text{gov}}$ for graph-conditioned diffusion in Gaussian parameter space, with a governance loss derived directly from the hypergraph rule layer and a hard-constraint projection operator for rule nodes that admit no soft violation.

5. **Community validation loop (Section 7).** A formal treatment of hypergraph update operators and an active learning criterion that prioritises annotation of high-uncertainty reconstruction regions.
6. **Field data collection protocol (Section 8).** A structured protocol for drone survey, historical imagery recovery, oral testimony recording, and community annotation session design, intended for on-site deployment in the High Atlas.

1.6 A note on prior formulations

The core equation $Y = H(S(Y'), G_m)$ introduced in earlier work on this problem, where Y is the generated design, Y' a preliminary output, S a weighted constraint sum, and G_m a flat governance metadata table, is a useful conceptual sketch. We show in Section 6 that its constrained-generation objective,

$$Y^* = \arg \max_{Y'} [-\log P(Y'|\cdot) + \lambda S(Y')], \quad (1)$$

is a special case of our more general formulation when (i) the governance hypergraph degenerates to a flat table, (ii) constraint interactions are neglected, and (iii) temporal and conflict structure are suppressed. Our contribution is to lift each of these restrictions in a mathematically precise way.

1.7 Organisation

Section 2 establishes the mathematical preliminaries: 3D and 4D Gaussian splatting, hypergraph theory, hypergraph neural networks, and constrained generative modelling. Sections 3 to 7 develop the five layers of the framework in sequence. Section 8 describes the on-site data collection protocol. Section 9 discusses open problems. Section 10 concludes.

2 Preliminaries

This section develops all mathematical background required for Sections 3 to 7. It is written for a reader who is technically strong but has not necessarily encountered formal mathematics as a discipline: we define every object, operator, and construction the first time it is used. ?? introduces the foundational vocabulary (sets, functions, vectors, matrices, groups, Lagrangians, graphs). Section 2.1 covers 3D Gaussian splatting. Section 2.2 is a self-contained primer on spherical harmonics. Section 2.3 covers the 4D extension. Section 2.4 covers hypergraph theory. Section 2.5 covers hypergraph neural networks. Section 2.6 covers constrained generative modelling.

2.1 Mathematical foundations

This subsection introduces the core mathematical vocabulary used throughout the paper.

2.1.1 Sets, functions, and notation

A *set* A is a collection of distinct objects, called its *elements*. We write $a \in A$ to mean that a is an element of A , and $a \notin A$ to mean it is not. The *empty set* $\emptyset$ contains no elements. The set of all subsets of A is the *power set* $\mathcal{P}(A) = \{B : B \subseteq A\}$; it has $2^{|A|}$ elements when A is finite.

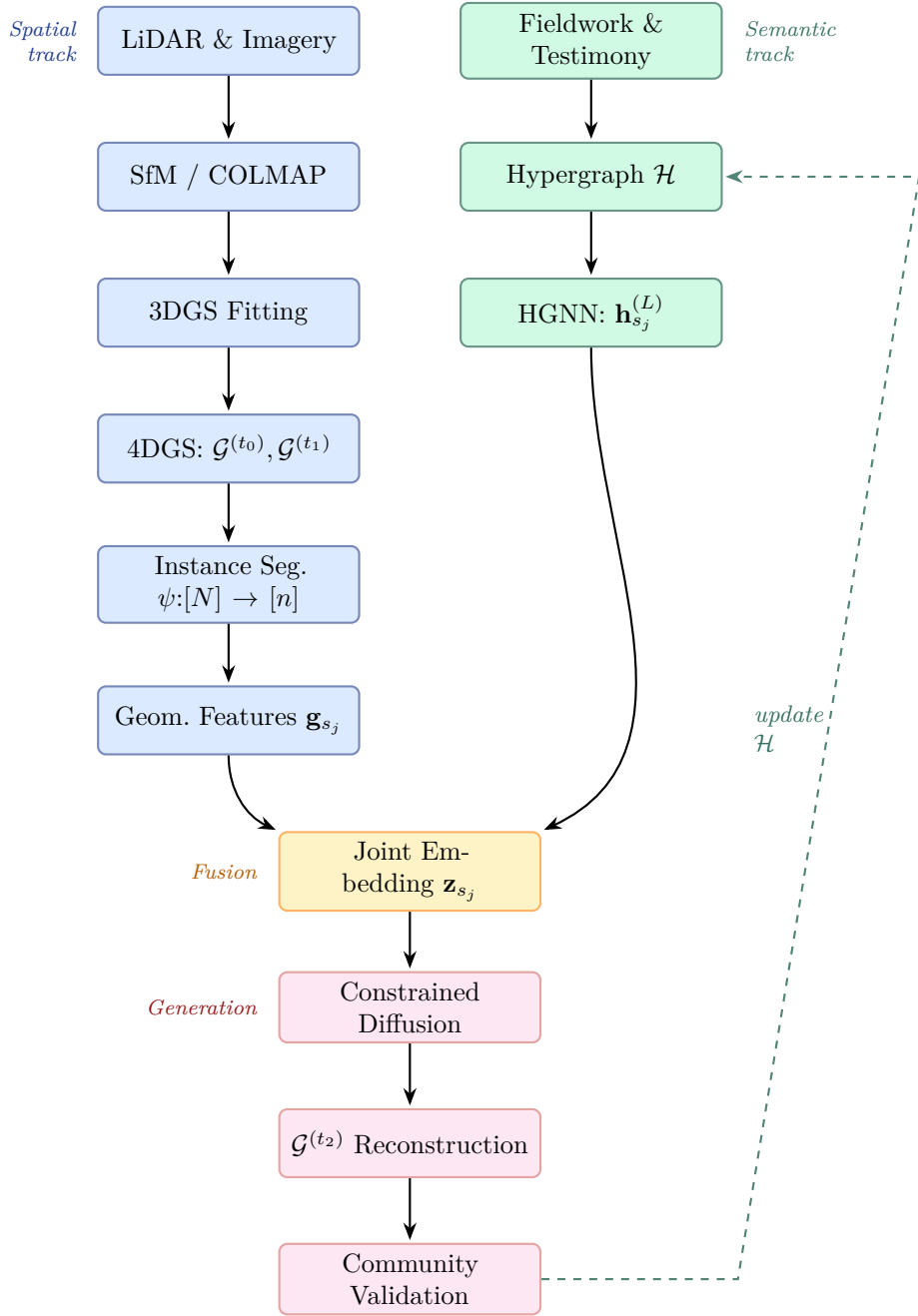


Figure 1: End-to-end system architecture (vertical layout). The *spatial track* (blue, left column) converts LiDAR and multi-view imagery into a 4D Gaussian scene and partitions it into structural instances with geometric feature vectors $\mathbf{g}_{s_j}$. The *semantic track* (green, right column) converts community fieldwork and oral testimony into the typed governance hypergraph $\mathcal{H}$, whose structural-vertex embeddings $\mathbf{h}_{s_j}^{(L)}$ are produced by the HGNN. The two tracks converge at the *fusion* stage (amber), which concatenates geometric and semantic features into the joint embedding $\mathbf{z}_{s_j}$. The *generation* stage (pink) runs constrained diffusion in Gaussian parameter space to produce the reconstruction proposal $\mathcal{G}^{(t_2)}$. Community validation closes the loop: stakeholder corrections produce hypergraph update operations (dashed arrow, right) that re-enter the semantic track.

Standard number sets: $\mathbb{N} = \{0, 1, 2, \dots\}$ (natural numbers), $\mathbb{Z}$ (integers), $\mathbb{Q}$ (rationals), $\mathbb{R}$ (real numbers), $\mathbb{R}_{>0}$ (positive reals). The notation $[n] = \{1, 2, \dots, n\}$ denotes the set of the first n positive integers.

A *function* $f : A \rightarrow B$ assigns to each element $a \in A$ exactly one element $f(a) \in B$. We call A the *domain* and B the *codomain*. A function is *injective* (one-to-one) if distinct inputs produce distinct outputs; *surjective* (onto) if every element of B is hit by some input; and *bijective* if both. The *composition* of $f : A \rightarrow B$ and $g : B \rightarrow C$ is $g \circ f : A \rightarrow C$, $(g \circ f)(a) = g(f(a))$.

The notation $\{x \in A : P(x)\}$ means the subset of A consisting of all x satisfying property P . The *intersection* $A \cap B$ contains elements in both A and B ; the *union* $A \cup B$ contains elements in either; the *difference* $A \setminus B$ contains elements of A not in B . The generalised intersection $\bigcap_i A_i$ is the set of elements belonging to every A_i simultaneously; it plays the role of “all constraints simultaneously satisfied” in Definition 6.3.

2.1.2 Vectors and the Euclidean space $\mathbb{R}^n$

A *vector* $\mathbf{x} \in \mathbb{R}^n$ is an ordered list of n real numbers $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$. The superscript $^\top$ (transpose) turns a row into a column; we write vectors as columns by convention. Addition of vectors and multiplication by scalars are component-wise: $\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_n + y_n)^\top$ and $\lambda \mathbf{x} = (\lambda x_1, \dots, \lambda x_n)^\top$.

The *dot product* (inner product) of two vectors is $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i \in \mathbb{R}$. The *Euclidean norm* is $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$, the length of the vector. A *unit vector* satisfies $\|\mathbf{x}\| = 1$.

2.1.3 Matrices, transposes, and multiplication

An $m \times n$ matrix $\mathbf{A}$ is a rectangular array of real numbers with m rows and n columns:

$$\mathbf{A} = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix}.$$

The *transpose* $\mathbf{A}^\top$ is the $n \times m$ matrix obtained by swapping rows and columns: $(\mathbf{A}^\top)_{ij} = A_{ji}$.

Matrix-vector multiplication: if $\mathbf{A}$ is $m \times n$ and $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{A}\mathbf{x} \in \mathbb{R}^m$ has i -th entry $\sum_{j=1}^n A_{ij}x_j$. This represents a *linear transformation*: $\mathbf{A}$ maps each input vector $\mathbf{x}$ to an output vector $\mathbf{A}\mathbf{x}$.

Matrix-matrix multiplication: if $\mathbf{A}$ is $m \times k$ and $\mathbf{B}$ is $k \times n$, then $\mathbf{C} = \mathbf{A}\mathbf{B}$ is $m \times n$ with entries $C_{ij} = \sum_{\ell=1}^k A_{i\ell}B_{\ell j}$. This is composition of linear transformations: applying $\mathbf{B}$ first, then $\mathbf{A}$.

A matrix $\mathbf{A}$ is *symmetric* if $\mathbf{A} = \mathbf{A}^\top$. It is *positive definite* (written $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for every non-zero $\mathbf{x}$; geometrically, it defines an ellipsoidal shape in $\mathbb{R}^n$. The set $\mathcal{S}_{++}^n$ denotes all $n \times n$ real symmetric positive-definite matrices. A symmetric matrix $\mathbf{A}$ can be decomposed as $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ (eigendecomposition), where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ holds the eigenvalues and $\mathbf{U}$ is orthogonal ($\mathbf{U}\mathbf{U}^\top = \mathbf{I}$).

The *identity matrix* $\mathbf{I}_n$ has ones on the diagonal and zeros elsewhere; it satisfies $\mathbf{I}\mathbf{x} = \mathbf{x}$ and $\mathbf{A}\mathbf{I} = \mathbf{I}\mathbf{A} = \mathbf{A}$. The *trace* of a square matrix is $\text{tr}(\mathbf{A}) = \sum_i A_{ii}$. The operator $\text{diag}(\mathbf{x})$ constructs the diagonal matrix with the entries of $\mathbf{x}$ on its diagonal; $\text{diag}(\mathbf{A})$ reads back the diagonal entries as a vector.

The *half-vectorisation* $\text{vech}(\mathbf{A})$ of a symmetric $n \times n$ matrix stacks the lower-triangular entries (including the diagonal) into a vector of length $n(n+1)/2$. It is used in Definition 3.7 to compactly represent the second-moment matrix of a structural instance.

2.1.4 Groups and the rotation group $\text{SO}(3)$

A *group* $(G, \cdot)$ is a set G equipped with an associative binary operation $\cdot$ that has an identity element e (satisfying $e \cdot g = g \cdot e = g$ for all g) and inverses ($g \cdot g^{-1} = e$ for every g). Groups formalise the notion of symmetry: rotations, reflections, and translations are all groups.

The *special orthogonal group* $\text{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} : \mathbf{R}\mathbf{R}^\top = \mathbf{I}, \det(\mathbf{R}) = 1\}$ is the group of rotations in three dimensions. Every $\mathbf{R} \in \text{SO}(3)$ is a 3×3 matrix that rotates vectors without stretching or reflecting them: $\|\mathbf{R}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$. The group operation is matrix multiplication; the inverse of $\mathbf{R}$ is $\mathbf{R}^\top$.

In practice, elements of $\text{SO}(3)$ are stored as *unit quaternions* $\mathbf{q} = (q_w, q_x, q_y, q_z)$ with $\|\mathbf{q}\| = 1$, because quaternions avoid the coordinate singularity of Euler angles and are more efficient to differentiate through during gradient-based optimisation.

The *special Euclidean group* $\text{SE}(3) = \text{SO}(3) \ltimes \mathbb{R}^3$ is the group of rigid-body motions (rotations plus translations). An element $(\mathbf{R}, \mathbf{t}) \in \text{SE}(3)$ transforms a point $\mathbf{x} \in \mathbb{R}^3$ as $\mathbf{x} \mapsto \mathbf{R}\mathbf{x} + \mathbf{t}$. This is the natural group of transformations for camera poses and scene alignments.

2.1.5 Optimisation, $\arg \min$, and $\arg \max$

Given a function $f : \mathcal{X} \rightarrow \mathbb{R}$, the *minimum value* is $\min_{x \in \mathcal{X}} f(x)$ and the *minimiser* (the point achieving it) is

$$\arg \min_{x \in \mathcal{X}} f(x) = \{x \in \mathcal{X} : f(x) \leq f(y) \ \forall y \in \mathcal{X}\}.$$

The analogous notation for maximisation is $\arg \max$. When the minimiser is unique, $\arg \min$ returns a single point; in general it is a set. In Definition 6.1, $\arg \min$ returns the reconstruction scene $\mathcal{G}'$ that minimises the total loss, treating all Gaussian parameters as decision variables.

The *penalty method* for constrained optimisation replaces a hard constraint $c(\mathbf{x}) \leq 0$ with an additional term in the objective: $\min_{\mathbf{x}} f(\mathbf{x}) + \lambda \max(0, c(\mathbf{x}))^2$. The parameter $\lambda \geq 0$ trades off between minimising f and satisfying the constraint: when λ is large, violations of $c(\mathbf{x}) \leq 0$ are heavily penalised. As $\lambda \rightarrow \infty$, solutions of the penalised problem converge to solutions of the original constrained problem.

2.1.6 The Lagrangian

The *Lagrangian* of a constrained optimisation problem

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad c_i(\mathbf{x}) = 0, \ i = 1, \dots, m$$

is the function $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^m \mu_i c_i(\mathbf{x})$, where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_m)$ are called *Lagrange multipliers*. The Lagrangian converts a constrained problem into an unconstrained one: at a solution, $\nabla_{\mathbf{x}} \mathcal{L} = 0$ and $c_i(\mathbf{x}) = 0$, expressing the fact that the gradient of f must be a linear combination of the gradients of the constraints. In Section 6.2, we note that the penalty-weighted total loss $\mathcal{L}_{\text{total}}$ is the Lagrangian relaxation of the hard-constrained reconstruction problem (16), with λ_1 and λ_2 playing the role of multipliers.

2.1.7 Probability and expectation

A *random variable* X takes values in some set $\mathcal{X}$ with a probability distribution. For continuous X with density $p(x)$, the *expectation* (mean value) is $\mathbb{E}[X] = \int x p(x) dx$, and the *variance* measures spread: $\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$. For a Gaussian random variable $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ in $\mathbb{R}^n$, $\mathbb{E}[X] = \boldsymbol{\mu}$ and $\text{Var}[X] = \boldsymbol{\Sigma}$ (the covariance matrix).

A *conditional distribution* $p(X|Y = y)$ describes the distribution of X given that Y takes the value y . Conditional distributions pervade the generative model of Section 6.3, where the reverse diffusion process is conditioned on the joint embedding $\mathbf{Z}$.

2.1.8 Graphs

A *graph* $G = (V, E)$ consists of a *vertex set* V (a finite set of nodes) and an *edge set* $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$ of unordered pairs of distinct vertices. An edge $\{u, v\}$ represents a pairwise relation between u and v : in a road network, vertices are intersections and edges are roads; in a social network, vertices are people and edges are friendships.

A *directed graph* has edges $E \subseteq V \times V$ that point from a source to a target. A *weighted graph* assigns a real number $w_e \in \mathbb{R}$ to each edge. A graph is *connected* if there is a path between any two vertices.

The *adjacency matrix* $\mathbf{A} \in \{0, 1\}^{|V| \times |V|}$ has $A_{uv} = 1$ if $\{u, v\} \in E$ and $A_{uv} = 0$ otherwise. The *degree* of vertex v is $d(v) = \sum_u A_{uv}$, the number of edges incident to v .

Graphs model *binary* relations. The governance relations in our framework are fundamentally *multi-way*: a single water-sharing rule links one channel, three terraces, and two households simultaneously. This motivates the generalisation to hypergraphs, introduced formally in Section 2.4.

2.2 Three-dimensional Gaussian splatting

We begin with the representation introduced by Kerbl et al. [6], which provides both the geometric backbone and the rendering mechanism for our framework.

Definition 2.1 (3D Gaussian). A *3D Gaussian* is a tuple

$$g = (\boldsymbol{\mu}, \boldsymbol{\Sigma}, \mathbf{c}, \alpha),$$

where $\boldsymbol{\mu} \in \mathbb{R}^3$ is the centre of mass in world-space coordinates; $\boldsymbol{\Sigma} \in \mathcal{S}_{++}^3$ (the cone of 3×3 real symmetric positive-definite matrices) is the spatial covariance, encoding the size and orientation of the Gaussian ellipsoid; $\mathbf{c} : \mathbb{S}^2 \rightarrow \mathbb{R}^3$ is a view-dependent colour function represented by a finite truncation of the spherical harmonic basis (explained in full in Section 2.2); and $\alpha \in (0, 1)$ is the opacity.

Remark 2.2 (World-space and the two spheres). The phrase *world-space* refers to the local Euclidean coordinate frame $\mathbb{R}^3$ established by the SfM reconstruction of the scene. Its origin is placed, by convention, at the centroid of the recovered sparse point cloud (roughly the centre of the reconstructed village), and its axes are determined by the bundle-adjustment optimisation, with no fixed orientation relative to geographic north or local vertical. Positions in this frame are measured in metres and span a few hundred metres at most.

Two distinct spheres $\mathbb{S}^2$ appear in the framework, and they must not be confused with each other or with the sphere of the Earth.

1. **The sphere of viewing directions.** The unit 2-sphere $\mathbb{S}^2 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$ appearing in $\mathbf{c} : \mathbb{S}^2 \rightarrow \mathbb{R}^3$ is the set of all unit vectors in the world-space frame $\mathbb{R}^3$. A point on this sphere represents the *direction* from which a Gaussian is observed by a camera. It is a 2-dimensional compact manifold whose geometry is entirely intrinsic to the local reconstruction coordinate frame.
2. **Earth’s sphere.** The geographic sphere on which the High Atlas villages actually reside is a sphere of radius $\approx 6,371$ km, parameterised by latitude and longitude. The villages are *points* on this sphere; their local tangent plane is the ambient $\mathbb{R}^3$ in which the reconstruction lives. The geographic sphere does not appear anywhere in the 3DGS or hypergraph mathematics; it enters only in the ground-control-point georeferencing of Section 8.1, where reconstruction coordinates are aligned to a geodetic datum (WGS 84).

Both spheres are topologically $\mathbb{S}^2$, but they operate at completely different spatial scales and serve completely different mathematical roles.

The covariance Σ is stored in factored form $\Sigma = \mathbf{R}\mathbf{S}\mathbf{S}^\top\mathbf{R}^\top$, where $\mathbf{R} \in \text{SO}(3)$ parameterises orientation as a unit quaternion and $\mathbf{S} = \text{diag}(s_1, s_2, s_3)$ with $s_i > 0$ parameterises scale. This factorisation ensures positive definiteness under unconstrained gradient descent.

Definition 2.3 (Gaussian scene). A *Gaussian scene* is a finite multi-set $\mathcal{G} = \{g_k\}_{k=1}^N$ of 3D Gaussians. Each element g_k has its own parameters $(\mu_k, \Sigma_k, \mathbf{c}_k, \alpha_k)$.

Rendering. For a camera with projection matrix $\mathbf{P} \in \mathbb{R}^{3 \times 4}$ and world-to-camera transform $\mathbf{W} \in \text{SE}(3)$, each Gaussian g_k is projected to a 2D Gaussian with mean $\mu_k^{2D} = \pi(\mathbf{P}\mathbf{W}\mu_k)$ and covariance $\Sigma_k^{2D} = \mathbf{J}\mathbf{W}\Sigma_k\mathbf{W}^\top\mathbf{J}^\top$, where $\mathbf{J}$ is the Jacobian of the projective transformation at $\mathbf{W}\mu_k$. Pixel colours are computed by alpha-compositing Gaussians sorted by increasing depth:

$$\hat{C}(p) = \sum_{k=1}^N \mathbf{c}_k(\mathbf{d}) \alpha_k \exp\left(-\frac{1}{2}(p - \mu_k^{2D})^\top (\Sigma_k^{2D})^{-1} (p - \mu_k^{2D})\right) \prod_{j < k} (1 - \alpha_j), \quad (2)$$

where $\mathbf{d}$ is the viewing direction at pixel p and products over $j < k$ accumulate transmittance over the sorted Gaussians in front of k .

Training objective. Given a set of posed input images $\{(I_i, \Pi_i)\}_{i=1}^M$ (image and camera pose), the scene $\mathcal{G}$ is optimised by minimising

$$\mathcal{L}_{\text{photo}} = \sum_{i=1}^M \left[(1 - \lambda_{\text{SSIM}}) \left\| I_i - \hat{I}_i(\mathcal{G}) \right\|_1 + \lambda_{\text{SSIM}} (1 - \text{SSIM}(I_i, \hat{I}_i(\mathcal{G}))) \right], \quad (3)$$

where $\hat{I}_i$ is rendered from pose Π_i via (2) and $\lambda_{\text{SSIM}} \approx 0.2$ in practice. Gaussians are adaptively densified and pruned during optimisation to allocate capacity where the photometric residual is largest.

2.3 Spherical harmonics and view-dependent colour

This subsection provides a self-contained introduction to spherical harmonics for readers approaching the topic for the first time. The material is necessary to understand why the colour function $\mathbf{c} : \mathbb{S}^2 \rightarrow \mathbb{R}^3$ in Definition 2.1 takes the form it does, and why the specific sphere $\mathbb{S}^2$

that appears here is the sphere of viewing directions in local world-space, not any geographic sphere.

Why view-dependent colour? A physical surface does not emit or reflect a single colour uniformly in all directions. The plastered earthen walls of a High Atlas kasbah appear warm ochre on the sunlit south face, cool grey in shadow, and have a faint specular sheen at glancing angles where the mica in the clay catches the light. A bidirectional reflectance distribution function (BRDF) captures this: it assigns to each incoming light direction and outgoing viewing direction a reflected radiance. In the 3DGS setting, the incoming light is not modelled explicitly; instead, each Gaussian stores a *view-dependent colour function* $\mathbf{c}(\mathbf{d})$ that directly maps the viewing direction $\mathbf{d} \in \mathbb{S}^2$ to an RGB colour response. If a single constant colour were stored per Gaussian, the rendered image would be flat and viewpoint-independent. The spherical harmonic parameterisation provides a compact and differentiable family of functions on $\mathbb{S}^2$ that can represent realistic view-dependent appearance at low computational cost.

Functions on the unit sphere. A *viewing direction* is a unit vector $\mathbf{d} \in \mathbb{R}^3$ with $\|\mathbf{d}\| = 1$. The set of all such vectors is the unit 2-sphere $\mathbb{S}^2 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$, a compact 2-dimensional Riemannian manifold. Every point on $\mathbb{S}^2$ can be parameterised by spherical coordinates (θ, φ) with polar angle $\theta \in [0, \pi]$ and azimuthal angle $\varphi \in [0, 2\pi)$ via the embedding

$$\mathbf{d} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta).$$

We want to represent an arbitrary square-integrable function $f : \mathbb{S}^2 \rightarrow \mathbb{R}$ as an infinite series. To do this, we need an orthonormal basis for the Hilbert space $L^2(\mathbb{S}^2)$ of square-integrable functions on the sphere, equipped with the inner product

$$\langle f, g \rangle = \int_{\mathbb{S}^2} f(\mathbf{d}) g(\mathbf{d}) d\omega,$$

where $d\omega = \sin \theta d\theta d\varphi$ is the standard area element on $\mathbb{S}^2$.

The Laplace–Beltrami operator. Just as the Fourier basis on the circle $\mathbb{S}^1$ consists of the eigenfunctions of the ordinary second-derivative operator $\frac{d^2}{d\theta^2}$, the natural basis for $L^2(\mathbb{S}^2)$ consists of the eigenfunctions of the *Laplace–Beltrami operator* $\Delta_{\mathbb{S}^2}$, the intrinsic Laplacian of the sphere. In spherical coordinates,

$$\Delta_{\mathbb{S}^2} f = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2}. \quad (4)$$

This operator arises naturally in the Helmholtz equation on $\mathbb{R}^3$ when one separates variables in spherical coordinates.

Spherical harmonics as eigenfunctions.

Definition 2.4 (Spherical harmonics). The *real spherical harmonics* $Y_\ell^m : \mathbb{S}^2 \rightarrow \mathbb{R}$, indexed by non-negative integer degree $\ell \geq 0$ and integer order m with $-\ell \leq m \leq \ell$, are the real-valued eigenfunctions of $\Delta_{\mathbb{S}^2}$:

$$\Delta_{\mathbb{S}^2} Y_\ell^m = -\ell(\ell + 1) Y_\ell^m. \quad (5)$$

They form a complete orthonormal system for $L^2(\mathbb{S}^2)$:

$$\langle Y_\ell^m, Y_{\ell'}^{m'} \rangle = \delta_{\ell\ell'} \delta_{mm'}.$$

For each degree ℓ , there are $2\ell + 1$ harmonics (one for each allowed value of m). Up to degree L , there are $(L + 1)^2$ harmonics in total. The eigenvalue $-\ell(\ell + 1)$ is the analogue of $-n^2$ in the Fourier case: higher ℓ corresponds to higher angular frequency, i.e. more rapid oscillation over the sphere's surface.

Explicit formulas. The real spherical harmonics are constructed from the associated Legendre polynomials $P_\ell^{|m|}$ as follows. Let $K_\ell^m = \sqrt{\frac{(2\ell+1)}{4\pi} \frac{(\ell-|m|)!}{(\ell+|m|)!}}$ be a normalisation constant. Then:

$$Y_\ell^m(\theta, \varphi) = \begin{cases} \sqrt{2} K_\ell^m \cos(m\varphi) P_\ell^m(\cos \theta) & \text{if } m > 0, \\ K_\ell^0 P_\ell^0(\cos \theta) & \text{if } m = 0, \\ \sqrt{2} K_\ell^{|m|} \sin(|m|\varphi) P_\ell^{|m|}(\cos \theta) & \text{if } m < 0. \end{cases}$$

The first four degrees are listed explicitly in Table 1. The degree-0 harmonic Y_0^0 is a positive constant: a Gaussian whose colour is described by Y_0^0 alone has the same colour from every viewing direction. The degree-1 harmonics are linear functions of the Cartesian coordinates (x, y, z) evaluated on $\mathbb{S}^2$, and describe a smooth directional gradient in colour (the difference between the sun-facing and shadow-facing sides of a wall). Degree-2 harmonics describe quadratic angular variation (such as highlights at oblique angles), and degree-3 harmonics capture cubic oscillation.

Table 1: Explicit real spherical harmonics up to degree $\ell = 2$. Here $x = \sin \theta \cos \varphi$, $y = \sin \theta \sin \varphi$, $z = \cos \theta$ are the Cartesian coordinates of $\mathbf{d} \in \mathbb{S}^2$, and all expressions are evaluated at $\mathbf{d}$.

Degree ℓ	Order m	$Y_\ell^m(\mathbf{d})$
0	0	$\frac{1}{2} \sqrt{\frac{1}{\pi}}$
1	-1	$\frac{1}{2} \sqrt{\frac{3}{\pi}} y$
1	0	$\frac{1}{2} \sqrt{\frac{3}{\pi}} z$
1	+1	$\frac{1}{2} \sqrt{\frac{3}{\pi}} x$
2	-2	$\frac{1}{2} \sqrt{\frac{15}{\pi}} xy$
2	-1	$\frac{1}{2} \sqrt{\frac{15}{\pi}} yz$
2	0	$\frac{1}{4} \sqrt{\frac{5}{\pi}} (2z^2 - x^2 - y^2)$
2	+1	$\frac{1}{2} \sqrt{\frac{15}{\pi}} xz$
2	+2	$\frac{1}{4} \sqrt{\frac{15}{\pi}} (x^2 - y^2)$

The SH expansion and truncation. Any $f \in L^2(\mathbb{S}^2)$ admits the expansion

$$f(\mathbf{d}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} c_\ell^m Y_\ell^m(\mathbf{d}), \quad c_\ell^m = \int_{\mathbb{S}^2} f(\mathbf{d}) Y_\ell^m(\mathbf{d}) d\omega. \quad (6)$$

This is the spherical analogue of a Fourier series. Truncating at degree L gives the best L^2 approximation to f by a linear combination of at most $(L + 1)^2$ basis functions:

$$f_L(\mathbf{d}) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} c_{\ell}^m Y_{\ell}^m(\mathbf{d}).$$

The approximation error $\|f - f_L\|_{L^2}$ decays as $L \rightarrow \infty$; the rate of decay is governed by the smoothness of f on the sphere.

In 3DGS, the colour function of each Gaussian is truncated at $L = 3$, giving $(3 + 1)^2 = 16$ basis functions per colour channel. The choice $L = 3$ reflects an empirical observation about natural materials: nearly all view-dependent variation in earthen, stone, and plaster surfaces is captured by low-degree harmonics. A perfectly diffuse (Lambertian) surface has colour $\mathbf{c}(\mathbf{d}) = c$ constant in $\mathbf{d}$, which is exactly described by Y_0^0 alone ($L = 0$). A surface with a directional shading gradient (dark on one side, light on the other) is well approximated by $L = 1$. Mirror-like specular highlights, which would require very high L , are absent in the High Atlas earthen-construction materials that form the primary subject of this framework.

Storage and evaluation in 3DGS. Each Gaussian g_k stores, for each colour channel $c \in \{R, G, B\}$, a vector of $(L + 1)^2 = 16$ real coefficients $\{c_{k,\ell,m}^{(c)}\}_{\ell=0,m=-\ell}^{L,\ell}$. Given a query viewing direction $\mathbf{d}$, the colour is evaluated as

$$\mathbf{c}_k(\mathbf{d}) = \left[\sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} c_{k,\ell,m}^{(c)} Y_{\ell}^m(\mathbf{d}) \right]_{c \in \{R, G, B\}}. \quad (7)$$

The direction $\mathbf{d}$ used in this evaluation is the *viewing direction in world-space*: the unit vector pointing from the Gaussian’s centre $\boldsymbol{\mu}_k$ toward the camera. It is computed in the local reconstruction coordinate frame and has no relation to any geographic direction. The SH coefficients $c_{k,\ell,m}^{(c)}$ are optimised alongside $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \alpha_k$ during the photometric training of Section 2.1.

Key insight

The key intuition for a reader new to spherical harmonics: they are to functions on the sphere what the Fourier basis $\{1, \sin \theta, \cos \theta, \sin 2\theta, \dots\}$ is to functions on the circle. The degree ℓ controls the angular frequency of the basis function, exactly as n controls the temporal frequency in $e^{in\theta}$. Storing 16 SH coefficients per Gaussian (and per colour channel) is analogous to storing the first 16 Fourier coefficients of a periodic signal: it captures the smooth low-frequency variation in colour with direction, while ignoring high-frequency detail that is imperceptible or absent in natural materials.

2.4 The four-dimensional extension

Static 3DGS cannot represent a scene that changes over time. Wu et al. [7] and Yang et al. [8] extend the model by endowing each Gaussian with a deformation field.

Definition 2.5 (4D Gaussian scene). A *4D Gaussian scene* augments each Gaussian g_k with a deformation $\mathbf{d}_k : \mathcal{T} \rightarrow \mathbb{R}^3$ and a rotation update $\Delta \mathbf{R}_k : \mathcal{T} \rightarrow \text{SO}(3)$, where $\mathcal{T} \subset \mathbb{R}$ is the time

domain. The time-dependent parameters are

$$\boldsymbol{\mu}_k(t) = \boldsymbol{\mu}_k + \mathbf{d}_k(t), \quad \mathbf{R}_k(t) = \Delta \mathbf{R}_k(t) \cdot \mathbf{R}_k.$$

In continuous-video settings $\mathbf{d}_k(t)$ is parameterised by a neural deformation network $f_\theta(t, k)$. In our framework $\mathcal{T}$ is discrete: the temporal domain reduces to three keyframes $\mathcal{T}_{\text{kf}} = \{t_0, t_1, t_2\}$ representing three distinct physical states of the scene, so $\mathbf{d}_k(t)$ is a lookup table with three entries per Gaussian. We discuss the structure of these keyframes in Section 3.3.

2.5 Hypergraph theory

Ordinary graphs represent pairwise relations. Governance rules over High Atlas landscapes are fundamentally multi-way: a single Agdal water-sharing norm constrains an irrigation channel, multiple terraced plots, and multiple household lineages simultaneously. A hypergraph is the natural mathematical setting.

Definition 2.6 (Hypergraph). A *hypergraph* is a pair $H = (V, \mathcal{E})$ where V is a finite vertex set and $\mathcal{E} \subseteq \mathcal{P}(V) \setminus \{\emptyset\}$ is a family of non-empty subsets of V , called *hyperedges*. An ordinary graph is the special case in which every hyperedge has cardinality exactly two.

Definition 2.7 (Incidence matrix). The *incidence matrix* of $H = (V, \mathcal{E})$ is the matrix $\mathbf{B} \in \{0, 1\}^{|V| \times |\mathcal{E}|}$ defined by

$$B_{ve} = \begin{cases} 1 & \text{if } v \in e, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.8 (Degrees). The *degree* of a vertex v is $d(v) = \sum_{e \in \mathcal{E}} B_{ve}$, the number of hyperedges containing v . The *degree* of a hyperedge e is $\delta(e) = |e|$, its cardinality. We write $\mathbf{D}_V = \text{diag}(d(v))_{v \in V}$ and $\mathbf{D}_E = \text{diag}(\delta(e))_{e \in \mathcal{E}}$ for the corresponding diagonal degree matrices.

Definition 2.9 (Hypergraph Laplacian). Let $\mathbf{W} = \text{diag}(w_e)_{e \in \mathcal{E}}$ assign a positive weight to each hyperedge. The *normalised hypergraph Laplacian* is

$$\Delta = \mathbf{I} - \mathbf{D}_V^{-1/2} \mathbf{B} \mathbf{W} \mathbf{D}_E^{-1} \mathbf{B}^\top \mathbf{D}_V^{-1/2}.$$

This generalises the Cheeger Laplacian of an ordinary graph and was introduced by Zhou et al. [16].

Definition 2.10 (Typed hypergraph). A *typed hypergraph* is a tuple $(V, \mathcal{E}, \phi_V, \phi_E)$ where $\phi_V : V \rightarrow \mathcal{T}_V$ and $\phi_E : \mathcal{E} \rightarrow \mathcal{T}_E$ assign a *type* to each vertex and hyperedge respectively, for finite type sets $\mathcal{T}_V$ and $\mathcal{T}_E$.

The type functions allow the relational structure to distinguish, for example, the edge “actor a owns structure s ” from the edge “rule r restricts structure s during period $[t^-, t^+]$ ”, even though both are hyperedges involving a (or r) and s . This distinction matters for message passing, as we see in Section 2.5.

2.6 Hypergraph neural networks

A hypergraph neural network (HGNN) propagates information through the incidence structure of a hypergraph to compute per-vertex representations that aggregate both local and global context.

Two-stage message passing. The standard HGNN layer performs message passing in two stages. Let $\mathbf{h}_v^{(\ell)} \in \mathbb{R}^{d_\ell}$ denote the embedding of vertex v at layer ℓ .

Stage 1 (vertex-to-hyperedge aggregation). For each hyperedge $e \in \mathcal{E}$, compute a hyperedge representation by aggregating over its member vertices:

$$\mathbf{z}_e^{(\ell)} = \rho_1\left(\{\mathbf{h}_v^{(\ell)} : v \in e\}\right),$$

where ρ_1 is a permutation-invariant aggregation function (sum, mean, or a learned multiset function).

Stage 2 (hyperedge-to-vertex aggregation). For each vertex v , aggregate the representations of all hyperedges containing v :

$$\mathbf{h}_v^{(\ell+1)} = \sigma\left(\mathbf{W}^{(\ell)} \rho_2\left(\{\mathbf{z}_e^{(\ell)} : v \in e\}\right)\right),$$

where $\mathbf{W}^{(\ell)} \in \mathbb{R}^{d_{\ell+1} \times d_\ell}$ is a learnable weight matrix and σ is a non-linearity.

The AllSet formulation. Chien et al. [13] show that both aggregations should be learned multiset functions to avoid information loss. In AllSet, ρ_1 and ρ_2 are each parameterised as

$$\rho(\{x_i\}) = \text{MLP}\left(\frac{1}{|\{x_i\}|} \sum_i \text{MLP}(x_i)\right),$$

which is a universal approximator over multisets. AllSet strictly subsumes the simpler hypergraph convolution of Feng et al. [12] and the attention variant of Bai et al. [17].

Type-sensitive message passing. For a typed hypergraph, we use separate weight matrices and aggregation functions for each hyperedge type $\tau \in \mathcal{T}_E$:

$$\mathbf{h}_v^{(\ell+1)} = \sigma\left(\sum_{\tau \in \mathcal{T}_E} \mathbf{W}_\tau^{(\ell)} \rho_{2,\tau}\left(\{\mathbf{z}_e^{(\ell)} : v \in e, \phi_E(e) = \tau\}\right)\right). \quad (8)$$

This is the relational generalisation of R-GCN [11] to the hypergraph setting.

2.7 Constrained generative modelling

We briefly review the denoising diffusion framework [14] and the role of penalty-based constraint enforcement.

Diffusion models. A diffusion model defines a forward process that gradually corrupts a data sample $x^{(0)}$ over T steps:

$$q(x^{(1:T)}|x^{(0)}) = \prod_{t=1}^T q(x^{(t)}|x^{(t-1)}), \quad q(x^{(t)}|x^{(t-1)}) = \mathcal{N}(x^{(t)}; \sqrt{1 - \beta_t} x^{(t-1)}, \beta_t \mathbf{I}),$$

and a reverse process p_θ learned to undo the corruption:

$$p_\theta(x^{(0:T)}) = p(x^{(T)}) \prod_{t=1}^T p_\theta(x^{(t-1)}|x^{(t)}).$$

The network $\epsilon_\theta(x^{(t)}, t)$ is trained to predict the noise ϵ added at step t via $\mathcal{L}_{\text{DDPM}} = \mathbb{E}[\|\epsilon - \epsilon_\theta(x^{(t)}, t)\|^2]$.

Conditioned generation. To generate samples consistent with a context $\mathbf{Z}$, the reverse process is conditioned: $p_\theta(x^{(t-1)}|x^{(t)}, \mathbf{Z})$. The conditioning signal $\mathbf{Z}$ enters via cross-attention in the denoising network.

Constraint enforcement. Two standard mechanisms enforce constraints during generation. Soft enforcement adds a constraint penalty $\lambda c(x)$ to the training loss. Hard enforcement applies a projection operator $\Pi_C : x \mapsto \arg \min_{x' \in C} \|x' - x\|$ after each denoising step, where C is the feasible set. Hard enforcement guarantees constraint satisfaction at the cost of potentially disrupting the diffusion trajectory; soft enforcement is differentiable but provides no guarantee. We use both in Section 6.

3 Layer I: The 4D Gaussian Scene

This section formalises the spatial layer of our framework: the geometric representation of the High Atlas landscape as a 4D Gaussian splat scene over three temporal keyframes.

3.1 Scene parameterisation

Let $\mathcal{T}_{\text{kf}} = (t_0, t_1, t_2)$ be an ordered triple of temporal keyframes with the following interpretations.

- t_0 : the *pre-earthquake* state, representing the landscape as it existed before 8 September 2023. The principal data source for this keyframe is archival and community photography.
- t_1 : the *post-earthquake damaged* state, derived from systematic drone survey imagery collected on site.
- t_2 : the *proposed reconstruction* state, produced by the generative layer (Section 6).

Definition 3.1 (Temporal Gaussian scene). For each keyframe $t \in \mathcal{T}_{\text{kf}}$, the *temporal Gaussian scene* is

$$\mathcal{G}^{(t)} = \{g_k^{(t)}\}_{k=1}^N, \quad g_k^{(t)} = (\boldsymbol{\mu}_k^{(t)}, \boldsymbol{\Sigma}_k^{(t)}, \mathbf{c}_k^{(t)}, \alpha_k^{(t)}),$$

where the parameters at keyframe t are related to a canonical reference by

$$\boldsymbol{\mu}_k^{(t)} = \boldsymbol{\mu}_k + \mathbf{d}_k(t), \quad \mathbf{R}_k^{(t)} = \Delta \mathbf{R}_k(t) \cdot \mathbf{R}_k.$$

The deformation $\mathbf{d}_k(t) \in \mathbb{R}^3$ and rotation update $\Delta \mathbf{R}_k(t) \in \text{SO}(3)$ are stored as three-entry lookup tables indexed by $t \in \mathcal{T}_{\text{kf}}$.

Remark 3.2. We deliberately avoid a continuous neural deformation field for the temporal axis. The keyframes t_0 and t_1 represent physically distinct states separated by a seismic event; interpolating between them is geophysically meaningless. A lookup table is both simpler and more faithful to the data-generating process.

3.2 Multi-view reconstruction

The scenes $\mathcal{G}^{(t_0)}$ and $\mathcal{G}^{(t_1)}$ are estimated from collections of posed photographs via the pipeline: structure-from-motion (SfM) to recover camera poses, followed by Gaussian fitting to minimise the photometric loss (3).

Pose recovery. Camera poses are recovered from unordered image collections using COLMAP [18], which solves the incremental SfM problem: extract and match SIFT features across images, recover a sparse 3D point cloud via bundle adjustment, and estimate intrinsic and extrinsic parameters for each camera. For archival and community photographs contributing to $\mathcal{G}^{(t_0)}$, the images typically have unknown camera metadata; COLMAP handles this via focal-length marginalisation and radial distortion estimation [19].

Gaussian fitting. Given poses $\{\Pi_i^{(t)}\}_{i=1}^{M_t}$ and corresponding images $\{I_i^{(t)}\}$, the scene $\mathcal{G}^{(t)}$ is initialised from the COLMAP sparse point cloud (one Gaussian per sparse point, identity covariance, uniform opacity) and then optimised by stochastic gradient descent on (3) with adaptive densification.

Keyframe registration. To enable deformation-field computation, $\mathcal{G}^{(t_0)}$ and $\mathcal{G}^{(t_1)}$ must be expressed in a common world-coordinate frame. Let $\mathbf{T}^* \in \text{SE}(3)$ be the rigid alignment minimising the Chamfer distance between the t_0 and t_1 mean clouds:

$$\mathbf{T}^* = \arg \min_{\mathbf{T} \in \text{SE}(3)} \sum_{k=1}^N \min_j \left\| \boldsymbol{\mu}_k^{(t_0)} - \mathbf{T} \boldsymbol{\mu}_j^{(t_1)} \right\|^2. \quad (9)$$

In practice the damaged scene $\mathcal{G}^{(t_1)}$ will contain *additional* Gaussians representing rubble and collapse debris and *fewer* Gaussians in collapsed structural regions; the Chamfer objective is robust to this asymmetry. After alignment, all scenes share a common coordinate system and the deformations $\mathbf{d}_k(t_1) = \boldsymbol{\mu}_k^{(t_1)} - \boldsymbol{\mu}_k^{(t_0)}$ can be interpreted as displacement vectors encoding earthquake damage.

3.3 Temporal keyframe structure

Definition 3.3 (Displacement field). The *damage displacement field* is the map

$$\mathbf{D} : \mathcal{G}^{(t_0)} \rightarrow \mathbb{R}^3, \quad \mathbf{D}(g_k) = \boldsymbol{\mu}_k^{(t_1)} - \boldsymbol{\mu}_k^{(t_0)},$$

defined after solving (9). For Gaussians in structurally intact regions, $\mathbf{D}(g_k) \approx \mathbf{0}$. Persistent large-magnitude displacements indicate collapse or debris accumulation.

Remark 3.4. The keyframe t_2 is not given but generated: it is the output of the constrained generation stage in Section 6. Its deformation $\mathbf{d}_k(t_2)$ represents the proposed reconstruction displacement from the reference configuration, subject to physical and governance constraints.

3.4 Gaussian instance segmentation

A raw Gaussian scene $\mathcal{G}$ is an unstructured collection of primitives. To reason about governance at the level of architectural elements, we must partition $\mathcal{G}$ into semantically coherent structural instances.

Definition 3.5 (Structural instance). A *structural instance* s_j is a maximal set of Gaussians that correspond to a single semantically coherent physical element: a building, a terrace, a water channel, a threshold, a granary, or a sacred enclosure. The set $\mathcal{S} = \{s_1, \dots, s_n\}$ partitions the physical elements of the scene.

Definition 3.6 (Instance assignment map). The *instance assignment map* is a surjection $\psi : [N] \rightarrow [n]$ such that $\psi(k) = j$ if and only if Gaussian g_k is assigned to instance s_j . The assignment induces a partition $\{G_{s_j}\}_{j=1}^n$ of $[N]$ by

$$\mathcal{G}_{s_j} = \{g_k \in \mathcal{G} : \psi(k) = j\}.$$

The instance map ψ is computed in two stages. First, a foundation segmentation model [9] produces per-Gaussian semantic label probabilities by projecting each Gaussian mean into multiple rendered views and aggregating the resulting masks. Second, Gaussians with the same dominant semantic label that are spatially adjacent (within a threshold ϵ in world space) are merged into a single instance. In highly damaged scenes, this process requires manual verification of a small subset of instances; we discuss the annotation interface for this in Section 7.

Recent work on Gaussian Grouping [20] provides a differentiable variant in which the instance labels are trained jointly with the Gaussian parameters, yielding sharper boundaries between instances. We treat this as an available implementation option; the formal definition of ψ is independent of the specific algorithm used to compute it.

3.5 Structural instance geometry

For each structural instance s_j , we compute a compact geometric feature vector that summarises the spatial and photometric properties of its constituent Gaussians.

Definition 3.7 (Geometric feature). The *geometric feature* of instance s_j is the vector $\mathbf{g}_{s_j} \in \mathbb{R}^{d_g}$ defined by

$$\begin{aligned} \bar{\boldsymbol{\mu}}_{s_j} &= \frac{\sum_{k:\psi(k)=j} \alpha_k \boldsymbol{\mu}_k}{\sum_{k:\psi(k)=j} \alpha_k} && \text{(opacity-weighted centroid),} \\ \mathbf{M}_{s_j} &= \frac{\sum_{k:\psi(k)=j} \alpha_k (\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}}_{s_j})(\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}}_{s_j})^\top}{\sum_{k:\psi(k)=j} \alpha_k} && \text{(second-moment matrix),} \\ v_{s_j} &= \sum_{k:\psi(k)=j} \alpha_k && \text{(total opacity),} \\ \bar{\mathbf{c}}_{s_j} &= \frac{1}{|\mathcal{G}_{s_j}|} \sum_{k:\psi(k)=j} \mathbf{c}_k && \text{(mean colour coefficients).} \end{aligned}$$

The feature vector is then $\mathbf{g}_{s_j} = [\bar{\boldsymbol{\mu}}_{s_j}^\top; \text{vech}(\mathbf{M}_{s_j}); v_{s_j}; \bar{\mathbf{c}}_{s_j}^\top]^\top$, where vech denotes the half-vectorisation of the symmetric matrix.

Remark 3.8. The opacity weighting in the centroid and second-moment computations ensures that lightly contributing Gaussians (those on the periphery of an instance, often with small α_k) do not unduly influence the geometric summary. The second-moment matrix $\mathbf{M}_{s_j}$ encodes the extent and orientation of the instance as an ellipsoid, and its eigenvalue decomposition provides a natural coordinate frame for the structural element.

4 Layer II: The Governance Hypergraph

The governance hypergraph $\mathcal{H} = (V, \mathcal{E}, \phi_V, \phi_E)$ is the formal representation of the social, cultural, and normative context of the landscape. It encodes who governs what, under what rules, for what reasons, and with what history. This section defines the vertex and hyperedge taxonomies,

derives the incidence structure, treats temporal governance, and establishes the necessity of conflict-preserving representation.

4.1 Vertex taxonomy

The vertex set V is partitioned into five typed subsets:

$$V = \mathcal{S} \cup \mathcal{A} \cup \mathcal{N} \cup \mathcal{R} \cup \mathcal{V},$$

with vertex type function $\phi_V : V \rightarrow \mathcal{T}_V$, $\mathcal{T}_V = \{\text{Str}, \text{Act}, \text{Nar}, \text{Rul}, \text{Evt}\}$.

Definition 4.1 (Structural vertex). Each element $s_j \in \mathcal{S}$ corresponds directly to a structural instance in $\mathcal{S}$ from Definition 3.5: a building, terrace, water channel, granary, threshold, or sacred enclosure. Structural vertices are the primary locus of governance: all other vertex types interact with the landscape through edges anchored at one or more structural vertices.

Definition 4.2 (Actor vertex). Each element $a \in \mathcal{A}$ represents a governance-relevant agent: an individual person, a named family lineage (*tawmat* [تأومات]), a household cluster, or a collective institution such as a village council (*jmaâa* [جماعة]) or a water-sharing guild. Actor vertices carry attributes including agent type, approximate lineage depth (number of generations of documented use), and current residency status.

Definition 4.3 (Narrative vertex). Each element $n \in \mathcal{N}$ is a record unit: a transcribed oral testimony with speaker provenance, date of recording, and language (Tachelhit, Tamazight, Darija, or French); or an archival document (cadastral record, heritage inventory, NGO survey, photographic archive entry) with source and date. Narrative vertices do not themselves prescribe governance; they provide evidential grounding for the edges that do.

Definition 4.4 (Rule vertex). Each element $r \in \mathcal{R}$ encodes a normative proposition in a structured form:

$$r = (\tau_r, C_r, [t_r^-, t_r^+], \sigma_r, \mathcal{P}_r),$$

where $\tau_r \in \{\text{permission}, \text{prohibition}, \text{obligation}\}$ is the deontic modality; C_r is a precondition formula (a Boolean expression over vertex attributes and edge membership); $[t_r^-, t_r^+]$ is the temporal validity interval (possibly $(-\infty, +\infty)$ for permanent rules); $\sigma_r \in [0, 1]$ is a cultural sensitivity score assigned during community sessions; and $\mathcal{P}_r \subseteq \mathcal{A}$ is the set of actors with the authority to modify or revoke r .

Example 4.5. The Agdal restriction on grazing in the upper Alpine meadows of the Oukaïmeden is represented as a rule vertex r_{agdal} with deontic modality *prohibition*, precondition $C_r :=$ “the current date falls in the closed-Agdal period”, temporal validity interval [1 April, 15 July] (approximate for the High Atlas agdal), sensitivity $\sigma_r = 0.93$ (high), and authority $\mathcal{P}_r$ the village council of the relevant fraction.

Definition 4.6 (Event vertex). Each element $e_v \in \mathcal{V}$ records a historical episode that changed the semantic status of one or more structural vertices: the original construction of a building, a fire, an inheritance transfer, the September 2023 earthquake, or a prior reconstruction. Event vertices carry a timestamp and a description; they serve as anchors for temporal reasoning about provenance and prior state.

4.2 Hyperedge taxonomy

We define six hyperedge types, collected in $\mathcal{T}_E$. The key property distinguishing each type is the cardinality and type-composition of the incident vertices.

Definition 4.7 (Ownership hyperedge). $e_{\text{owns}} \subseteq \mathcal{P}(\mathcal{A} \cup \mathcal{S})$, with $e_{\text{owns}} \cap \mathcal{A} \neq \emptyset$ and $e_{\text{owns}} \cap \mathcal{S} \neq \emptyset$. A single ownership hyperedge encodes a governance claim: the actors in $e_{\text{owns}} \cap \mathcal{A}$ assert collective ownership or stewardship over the structures in $e_{\text{owns}} \cap \mathcal{S}$. Collective claims naturally require hyperedges of cardinality greater than two.

Definition 4.8 (Governance hyperedge). $e_{\text{gov}} \subseteq \mathcal{P}(\mathcal{R} \cup \mathcal{S} \cup \mathcal{A})$, with $e_{\text{gov}} \cap \mathcal{R} \neq \emptyset$. A governance hyperedge binds a rule $r \in \mathcal{R}$ to the spatial structures and actors it governs. This is the primary mechanism by which Agdal norms, water-turn schedules, and land-use restrictions are attached to spatial elements.

Definition 4.9 (Testimony hyperedge). $e_{\text{test}} \subseteq \mathcal{P}(\mathcal{N} \cup \mathcal{S} \cup \mathcal{V})$, with $e_{\text{test}} \cap \mathcal{N} \neq \emptyset$. A testimony hyperedge links a narrative record $n \in \mathcal{N}$ to the structural elements and historical events it describes, providing an evidential chain from oral or archival knowledge to the geometric layer.

Definition 4.10 (Co-functional hyperedge). $e_{\text{cofunc}} \subseteq \mathcal{P}(\mathcal{S})$, $|e_{\text{cofunc}}| \geq 2$. A co-functional hyperedge groups structural elements that constitute a single functional unit: a water channel together with all terraces it irrigates, or a granary together with the threshing floor and storage alcoves that compose it. Constraint violations by one member of a co-functional hyperedge propagate to all members.

Definition 4.11 (Contested hyperedge). $e_{\text{cont}} \subseteq \mathcal{P}(\mathcal{A} \cup \mathcal{S})$, satisfying the additional condition that there exist $e'_{\text{cont}}, e''_{\text{cont}} \in \mathcal{E}$ with $\phi_E(e') = \phi_E(e'') = \text{owns}$, $e' \cap \mathcal{S} = e'' \cap \mathcal{S} = \{s\}$ for some $s \in \mathcal{S}$, and $e' \cap \mathcal{A} \neq e'' \cap \mathcal{A}$. In plain terms: two or more distinct ownership hyperedges claim the same structure. We represent the *dispute itself* as an explicit contested hyperedge e_{cont} involving all disputing actors and the contested structure, with a **status** attribute $\in \{\text{active}, \text{mediation}, \text{resolved}\}$.

Definition 4.12 (Temporal hyperedge). $e_{\text{temp}} \subseteq \mathcal{P}(\mathcal{V} \cup \mathcal{S})$, augmented with a timestamp $\tau(e_{\text{temp}}) \in \mathbb{R}$. A temporal hyperedge records that an event $v_e \in \mathcal{V}$ changed the state of a set of structural elements at time τ . The sequence of temporal hyperedges incident to a structure s_j forms its *provenance chain*.

4.3 The incidence structure

Given the typed hypergraph $\mathcal{H}$, the incidence matrix $\mathbf{B}$ (Definition 2.7) can be decomposed by edge type. For each $\tau \in \mathcal{T}_E$, define the *type-restricted incidence matrix*

$$\mathbf{B}_{ve}^\tau = B_{ve} \cdot \mathbf{1}[\phi_E(e) = \tau],$$

so that $\mathbf{B} = \sum_{\tau \in \mathcal{T}_E} \mathbf{B}^\tau$. This decomposition underlies the type-sensitive HGNN of (8).

The *governance-restricted* incidence $\mathbf{B}^{\text{gov}}$ deserves special attention: its rows indexed by structural vertices $s_j \in \mathcal{S}$ encode which governance rules directly constrain each structure. For a structural vertex s_j ,

$$[\mathbf{B}^{\text{gov}}]_{s_j, \cdot} = \mathbf{1}[\exists r \in \mathcal{R} : (r \in e) \wedge (s_j \in e) \wedge (\phi_E(e) = \text{gov})].$$

This row vector is the primary input to the governance constraint loss in Section 6.2.

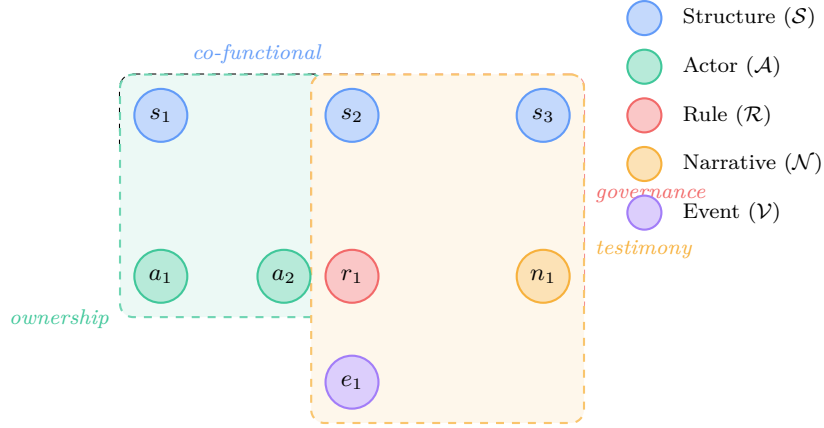


Figure 2: A fragment of the governance hypergraph $\mathcal{H}$. Structural vertices s_1, s_2, s_3 (blue) are linked by a co-functional hyperedge (the dashed box enclosing s_1, s_2). An ownership hyperedge binds actors a_1, a_2 and structure s_1 . A governance hyperedge binds rule r_1 to structures s_2, s_3 . A testimony hyperedge binds narrative n_1 to structure s_3 and event e_1 . Node colour encodes vertex type; the same structure node participates in multiple hyperedges simultaneously.

4.4 Temporal governance

Amazigh governance norms are not time-homogeneous. The Agdal system imposes seasonal restrictions; inheritance transfers modify ownership at discrete points in time; damage events alter the physical and normative status of structures.

Definition 4.13 (Temporally active rule). At real calendar time τ , a rule vertex $r \in \mathcal{R}$ is *active* if and only if

$$\tau \in [t_r^-, t_r^+] \quad \text{and} \quad C_r \text{ evaluates to True given the current state of } \mathcal{H}.$$

The set of active rules at time τ is $\mathcal{R}_\tau = \{r \in \mathcal{R} : r \text{ is active at } \tau\}$.

Definition 4.14 (Temporally active governance hyperedge). A governance hyperedge $e \in \mathcal{E}$ with $\phi_E(e) = \text{gov}$ is *active* at time τ if the rule vertex $r = e \cap \mathcal{R}$ (there is exactly one rule per governance hyperedge) belongs to $\mathcal{R}_\tau$.

Remark 4.15. The constraint layer in Section 6 evaluates governance compliance at the nominal reconstruction time τ_{rec} , which is set to the expected construction completion date. Rules whose validity intervals do not contain τ_{rec} are excluded from the constraint loss. Seasonal restrictions that overlap with the reconstruction period therefore impose actual constraints on the generated design.

4.5 Conflict-preserving representation

A central design principle of our framework is that contested ownership claims must be represented explicitly and preserved intact rather than collapsed into a single “resolved” state. We now make this principle precise and prove it necessary.

Definition 4.16 (Conflict pair). A *conflict pair* over structure $s_j \in \mathcal{S}$ is a pair of distinct ownership hyperedges $(e', e'') \in \mathcal{E}^2$ such that

- (i) $\phi_E(e') = \phi_E(e'') = \text{owns}$,

- (ii) $s_j \in e' \cap e''$,
- (iii) $e' \cap \mathcal{A} \neq e'' \cap \mathcal{A}$ (distinct claimant sets).

Definition 4.17 (Collapsed hypergraph). Given a hypergraph $\mathcal{H}$ containing a conflict pair (e', e'') over s_j , the *collapsed* hypergraph $\bar{\mathcal{H}}$ is obtained by removing e' and e'' and inserting a single ownership hyperedge $\bar{e}$ with $\bar{e} \cap \mathcal{A} = e' \cap \mathcal{A}$ (i.e., resolving in favour of one claimant). Collapsing in favour of e'' is defined analogously.

Proposition 4.18 (Necessity of conflict preservation). *Let $C_{\mathcal{H}}$ denote the set of governance-feasible reconstruction scenes for hypergraph $\mathcal{H}$, defined as*

$$C_{\mathcal{H}} = \{\mathcal{G}' : \mathcal{L}_{\text{gov}}(\mathcal{G}', \mathcal{H}) = 0\},$$

where $\mathcal{L}_{\text{gov}}$ is the governance constraint loss from Section 6. For any conflict pair (e', e'') in $\mathcal{H}$, let $\bar{\mathcal{H}}$ be the collapsed hypergraph favouring e' . Then in general $C_{\mathcal{H}} \subset C_{\bar{\mathcal{H}}}$, i.e., collapsing expands the feasible set, admitting designs that violate the original contested claim.

Proof. Under the collapsed hypergraph $\bar{\mathcal{H}}$, actor set $e'' \cap \mathcal{A}$ holds no ownership claim over s_j . Therefore any design that, under $\mathcal{H}$, would violate e'' 's claim (e.g., by constructing a wall that blocks access the e'' -claimant lineage historically enjoyed) is no longer recognised as a constraint violation in $C_{\bar{\mathcal{H}}}$. Formally, let $\mathcal{G}^*$ be any scene satisfying $\mathcal{L}_{\text{gov}}(\mathcal{G}^*, \mathcal{H}) = 0$. By the constraint structure, $\mathcal{G}^*$ must simultaneously respect both e' and e'' , so it belongs to $C_{\mathcal{H}}$. But there exist scenes $\mathcal{G}^{**}$ with $\mathcal{L}_{\text{gov}}(\mathcal{G}^{**}, \bar{\mathcal{H}}) = 0$ that violate e'' 's claim, hence $\mathcal{G}^{**} \notin C_{\mathcal{H}}$. Thus $C_{\mathcal{H}} \subsetneq C_{\bar{\mathcal{H}}}$. $\square$

Key insight

Proposition 4.18 formalises a key social fact: by collapsing a contested claim, the design system implicitly decides the dispute on behalf of the community. This is governance by technical default. Conflict-preserving representation ensures that the set of admissible reconstructions is the intersection of all plausible governance interpretations, not the union of one.

4.6 Formal properties of the governance hypergraph

We close this section with two structural properties that relate to the soundness and completeness of the governance encoding.

Definition 4.19 (Governance consistency). The hypergraph $\mathcal{H}$ is *governance-consistent* if no rule vertex $r \in \mathcal{R}$ is simultaneously active and its logical negation is implied by the current hyperedge structure. Informally, no pair of active governance edges simultaneously requires and prohibits the same spatial configuration.

Definition 4.20 (Structural coverage). The hypergraph $\mathcal{H}$ is *structurally complete* with respect to an instance set $\mathcal{S}$ if every structural vertex $s_j \in \mathcal{S}$ is incident to at least one ownership hyperedge and at least one testimony hyperedge. Structural completeness is the minimum standard required before committing to constrained generation.

Remark 4.21. In practice $\mathcal{H}$ will not be structurally complete after the first round of community annotation. The active learning criterion in Section 7 prioritises annotation of structurally incomplete vertices.

5 Layer III: Gaussian-to-Hypergraph Alignment

The spatial layer $\mathcal{G}$ and the semantic layer $\mathcal{H}$ are coupled through two operations: the instance assignment map ψ (Definition 3.6), which routes each Gaussian to a structural vertex, and the joint embedding, which fuses geometric and semantic information at each structural instance.

5.1 The instance alignment

The structural instance vertices $\mathcal{S} \subset V$ are in bijective correspondence with the structural instances $\{s_1, \dots, s_n\}$ of the spatial layer: each $s_j \in \mathcal{S}$ identifies both a hypergraph vertex and a set of Gaussians $\mathcal{G}_{s_j}$. The instance assignment map $\psi : [N] \rightarrow [n]$ (Definition 3.6) therefore induces a map $\tilde{\psi} : \mathcal{G} \rightarrow \mathcal{S}$ defined by $\tilde{\psi}(g_k) = s_{\psi(k)}$.

This coupling has two consequences. First, any attribute of a structural vertex s_j in the hypergraph (ownership claims, active rules, testimony records) is semantically associated with the set of Gaussians $\mathcal{G}_{s_j}$. Second, any physical modification of the Gaussians in $\mathcal{G}_{s_j}$ (in the generative stage) is subject to the governance constraints incident at s_j .

5.2 Geometric aggregation

Each structural vertex s_j receives a geometric feature vector $\mathbf{g}_{s_j}$ from Definition 3.7. For the post-earthquake keyframe t_1 , this feature vector is extended with a damage indicator:

$$\mathbf{g}_{s_j}^{(t_1)} = [\mathbf{g}_{s_j}; d_{s_j}; \bar{\mathbf{D}}_{s_j}]^\top,$$

where $d_{s_j} = \frac{1}{|\mathcal{G}_{s_j}|} \sum_{k:\psi(k)=j} \|\mathbf{D}(g_k)\|$ is the mean damage displacement magnitude for the instance, and $\bar{\mathbf{D}}_{s_j} = \frac{1}{|\mathcal{G}_{s_j}|} \sum_{k:\psi(k)=j} \mathbf{D}(g_k)$ is the mean damage displacement vector. Large d_{s_j} indicates collapse or severe displacement; near-zero d_{s_j} indicates structural survival.

5.3 Joint embedding

Definition 5.1 (Joint embedding). Let $\mathbf{h}_{s_j}^{(L)} \in \mathbb{R}^{d_h}$ be the HGNN output embedding of structural vertex s_j after L message-passing layers, computed via (8). Let $\mathbf{g}_{s_j} \in \mathbb{R}^{d_g}$ be the geometric feature of Definition 3.7. The *joint embedding* of instance s_j is

$$\mathbf{z}_{s_j} = \text{MLP}_\phi(\mathbf{W}_h \mathbf{h}_{s_j}^{(L)} \parallel \mathbf{W}_g \mathbf{g}_{s_j}) \in \mathbb{R}^{d_z}, \quad (10)$$

where $\mathbf{W}_h \in \mathbb{R}^{d_z/2 \times d_h}$ and $\mathbf{W}_g \in \mathbb{R}^{d_z/2 \times d_g}$ are linear projections, $\parallel$ denotes concatenation, and MLP_ϕ is a two-layer network with parameters ϕ . The matrix $\mathbf{Z} \in \mathbb{R}^{n \times d_z}$ stacks all joint embeddings row-wise.

Remark 5.2. The projection dimensions $d_z/2$ for each modality enforce a balanced fusion prior before the MLP. An alternative is to use a cross-attention mechanism over the Gaussian point cloud conditioned on the HGNN embeddings; we adopt the simpler concatenation scheme here and note the attention variant as an open architectural question (Section 9).

The joint embedding $\mathbf{z}_{s_j}$ now carries, for each structural instance: (i) its geometric shape, extent, opacity, and photometric character; (ii) the aggregate of all governance rules incident at s_j , weighted by hyperedge type; (iii) the collective testimonies and actor associations of s_j ; and (iv) the full second-order relational context via the multi-hop HGNN receptive field. This combined representation is the sole conditioning signal for the constrained generation stage.

6 Layer IV: Constrained Generative Reconstruction

Given the post-earthquake scene $\mathcal{G}^{(t_1)}$ and the joint embedding matrix $\mathbf{Z}$, the generation stage produces a proposed reconstruction $\mathcal{G}^{(t_2)}$ that is geometrically consistent with the pre-earthquake reference $\mathcal{G}^{(t_0)}$, physically sound, and governance-compliant with respect to $\mathcal{H}$.

6.1 Problem formulation

Definition 6.1 (Reconstruction problem). Let $\mathcal{G}^{(t_0)}$ and $\mathcal{G}^{(t_1)}$ be the pre- and post-earthquake scenes, $\mathcal{H}$ the governance hypergraph, and $\mathbf{Z}$ the joint embedding. The *governance-constrained reconstruction problem* is

$$\mathcal{G}^{(t_2)*} = \arg \min_{\mathcal{G}'} \mathcal{L}_{\text{total}}(\mathcal{G}', \mathcal{G}^{(t_0)}, \mathcal{H}),$$

where $\mathcal{L}_{\text{total}}$ is defined in Definition 6.2 below. The reconstruction $\mathcal{G}^{(t_2)*}$ is the proposed design submitted to community validation.

6.2 The reconstruction objective

Definition 6.2 (Total loss). The total loss is

$$\mathcal{L}_{\text{total}}(\mathcal{G}', \mathcal{G}^{(t_0)}, \mathcal{H}) = \mathcal{L}_{\text{recon}}(\mathcal{G}', \mathcal{G}^{(t_0)}) + \lambda_1 \mathcal{L}_{\text{struct}}(\mathcal{G}') + \lambda_2 \mathcal{L}_{\text{gov}}(\mathcal{G}', \mathcal{H}), \quad (11)$$

where $\lambda_1, \lambda_2 \geq 0$ are penalty weights and the three component losses are defined below.

Reconstruction fidelity loss. The reconstruction should be consistent with the pre-earthquake geometry:

$$\mathcal{L}_{\text{recon}}(\mathcal{G}', \mathcal{G}^{(t_0)}) = \sum_{j=1}^n w_j^{\text{rec}} \text{CD}(\mathcal{G}'_{s_j}, \mathcal{G}^{(t_0)}_{s_j}), \quad (12)$$

where CD is the bidirectional Chamfer distance between two Gaussian mean clouds and $w_j^{\text{rec}} \in [0, 1]$ is an instance-level fidelity weight. For structurally intact instances (small d_{s_j}), w_j^{rec} is set high; for fully collapsed instances with no surviving Gaussian geometry, w_j^{rec} is set low and the model relies on governance constraints and community annotation to guide reconstruction.

Structural feasibility loss. The reconstruction must respect seismic codes and structural physics. We encode a simplified set of structural constraints, operating on the instance bounding ellipsoids derived from $\mathbf{M}_{s_j}$:

$$\mathcal{L}_{\text{struct}}(\mathcal{G}') = \sum_{j=1}^n [\ell_{\text{floor}}(s_j) + \ell_{\text{clearance}}(s_j) + \ell_{\text{mass}}(s_j)], \quad (13)$$

where ℓ_{floor} penalises Gaussians placed below the terrain surface, $\ell_{\text{clearance}}$ penalises insufficient vertical clearance in habitable spaces, and ℓ_{mass} penalises volume-to-height ratios inconsistent with earthen construction load bearing. These are rough physical proxies; a complete structural analysis would require a finite-element model of the proposed design, which is beyond the scope of this framework.

Governance constraint loss. The governance loss measures violations of active rule vertices with respect to the proposed scene. For each active rule $r \in \mathcal{R}_{\tau}$, define the violation measure

$c_r : \mathcal{G}' \times \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ as follows. Rule r specifies a deontic modality τ_r and is incident to a set of structural vertices $\mathcal{S}_r = e_r \cap \mathcal{S}$ via its governance hyperedge e_r . The violation is:

$$c_r(\mathcal{G}', \mathcal{H}) = \begin{cases} \sum_{s_j \in \mathcal{S}_r} \max(0, f_r(\mathcal{G}'_{s_j}))^2 & \text{if } \tau_r = \text{prohibition,} \\ \sum_{s_j \in \mathcal{S}_r} \max(0, -f_r(\mathcal{G}'_{s_j}))^2 & \text{if } \tau_r = \text{obligation,} \end{cases} \quad (14)$$

where $f_r : \mathcal{G}'_{s_j} \rightarrow \mathbb{R}$ is a rule-specific *constraint function* that returns a positive value when the configuration of instance s_j violates the prohibition, and a negative value when the configuration fulfils an obligation. The governance loss aggregates over all active rules:

$$\mathcal{L}_{\text{gov}}(\mathcal{G}', \mathcal{H}) = \sum_{r \in \mathcal{R}_\tau} c_r(\mathcal{G}', \mathcal{H}). \quad (15)$$

Key insight

The naïve weighted-sum formulation $S(Y') = \sum_i w_i s_i(Y')$ of prior work aggregates constraint violations independently. Our formulation (15) preserves dependency structure by routing each rule’s violation through its governance hyperedge e_r : a co-functional hyperedge between s_1 and s_2 means that the constraint function f_r evaluating the irrigation channel s_1 also evaluates all terraces linked to it by a co-functional edge, not s_1 in isolation. The flat-sum approach is recovered if all co-functional hyperedges are removed.

The Lagrangian interpretation. The total loss (11) is an augmented penalty-method relaxation of the constrained programme

$$\min_{\mathcal{G}'} \mathcal{L}_{\text{recon}}(\mathcal{G}', \mathcal{G}^{(t_0)}) + \lambda_1 \mathcal{L}_{\text{struct}}(\mathcal{G}') \quad \text{subject to} \quad c_r(\mathcal{G}', \mathcal{H}) = 0 \quad \forall r \in \mathcal{R}_\tau. \quad (16)$$

As $\lambda_2 \rightarrow \infty$, any solution to the unconstrained problem (11) converges to a solution of (16). In practice λ_2 is tuned by community validation: if the community identifies governance violations in the proposed $\mathcal{G}^{(t_2)*}$, λ_2 is increased. If the proposed scene is governance-compliant but architecturally degraded, λ_2 is decreased.

6.3 Architecture: graph-conditioned diffusion in Gaussian space

We realise the generation as a denoising diffusion process that operates directly on Gaussian parameters [14, 15].

Gaussian parameter vector. For each instance s_j , flatten the parameters of all member Gaussians into a concatenated vector

$$\mathbf{x}_{s_j} = [\dots; \boldsymbol{\mu}_k; \text{vech}(\boldsymbol{\Sigma}_k); \alpha_k; \mathbf{c}_k; \dots]_{k \in \mathcal{G}_{s_j}},$$

and define $\mathbf{x} = [\mathbf{x}_{s_1}^\top; \dots; \mathbf{x}_{s_n}^\top]^\top$ as the full scene parameter vector. The generative model produces a proposal $\mathbf{x}^{(t_2)}$ from which $\mathcal{G}^{(t_2)}$ is trivially recovered.

Forward process. Starting from the post-earthquake parameters $\mathbf{x}^{(t_1)}$ (interpreted as the clean data $\mathbf{x}^{(0)}$ for the diffusion process), the forward process adds Gaussian noise over T diffusion steps:

$$q(\mathbf{x}^{(\tau)}|\mathbf{x}^{(0)}) = \mathcal{N}(\mathbf{x}^{(\tau)}; \sqrt{\bar{\alpha}_\tau} \mathbf{x}^{(0)}, (1 - \bar{\alpha}_\tau)\mathbf{I}), \quad (17)$$

where $\bar{\alpha}_\tau = \prod_{i=1}^\tau (1 - \beta_i)$ and $\{\beta_\tau\}$ is a noise schedule.

Conditioned reverse process. The reverse process is conditioned on $\mathbf{Z}$ and on the damage displacement field:

$$p_\theta(\mathbf{x}^{(\tau-1)}|\mathbf{x}^{(\tau)}, \mathbf{Z}, \mathbf{D}) = \mathcal{N}(\mathbf{x}^{(\tau-1)}; \boldsymbol{\mu}_\theta(\mathbf{x}^{(\tau)}, \mathbf{Z}, \mathbf{D}, \tau), \sigma_\tau^2 \mathbf{I}). \quad (18)$$

The denoising network $\epsilon_\theta(\mathbf{x}^{(\tau)}, \mathbf{Z}, \mathbf{D}, \tau)$ is a graph neural network that processes the Gaussian parameter vectors at all structural instances jointly, with cross-attention to the joint embedding matrix $\mathbf{Z}$ and the damage vector field $\mathbf{D}$. The GNN backbone propagates messages across the co-functional hyperedges of $\mathcal{H}$, ensuring that the denoising of one instance is informed by the state of its functionally coupled neighbours.

6.4 Hard constraint projection

Rule vertices of type *prohibition* with sensitivity $\sigma_r = 1$ (maximum cultural sensitivity) are treated as hard constraints that may not be violated at inference, regardless of the value of λ_2 . For these rules, we apply a projection operator $\Pi_{\mathcal{R}}$ after each denoising step.

Definition 6.3 (Hard constraint projection). Let $\mathcal{C}_r \subset \mathbb{R}^{|\mathbf{x}|}$ be the constraint-feasible set for hard rule r . The *rule projection operator* $\Pi_{\mathcal{R}_{\text{hard}}}$ is

$$\Pi_{\mathcal{R}_{\text{hard}}}(\mathbf{x}) = \arg \min_{\mathbf{x}' \in \bigcap_{r \in \mathcal{R}_{\text{hard}}} \mathcal{C}_r} \|\mathbf{x}' - \mathbf{x}\|,$$

applied after each reverse diffusion step. If the feasible sets $\{\mathcal{C}_r\}$ are all convex, the projection is unique and efficient; if they are not convex (the more common case), we use a sequential projection heuristic over the hard rules.

Remark 6.4. Introducing projection into the diffusion chain can disrupt the learned score and degrade sample quality. The severity depends on how often the unconstrained reverse process violates the hard constraints. In our setting, the most common hard constraints (sacred enclosures that must not be built upon; water channels whose alignment must not be altered) typically affect a small subset of structural instances, so the projection rarely fires and sample quality is minimally affected.

7 Layer V: The Community Validation Loop

The framework is not a one-shot pipeline. The proposed reconstruction $\mathcal{G}^{(t_2)*}$ is rendered and presented to community stakeholders, whose corrections are formalised as hypergraph update operations. These updates propagate back through the HGNN to refine the joint embeddings and trigger re-generation. This feedback loop is the primary mechanism by which field knowledge enters the formal system.

7.1 Hypergraph update operators

We define five atomic update operators on the hypergraph $\mathcal{H}$. Their composition describes all possible feedback actions.

Definition 7.1 (Hypergraph update operators). Let $\mathcal{H} = (V, \mathcal{E}, \phi_V, \phi_E)$.

$$\begin{aligned} \text{AddNode}(v, \tau_v, \mathbf{a}_v) &: \mathcal{H} \mapsto (V \cup \{v\}, \mathcal{E}, \phi_V[\tau_v], \phi_E), \\ \text{AddEdge}(e, \tau_e) &: \mathcal{H} \mapsto (V, \mathcal{E} \cup \{e\}, \phi_V, \phi_E[\tau_e]), \\ \text{RemoveEdge}(e) &: \mathcal{H} \mapsto (V, \mathcal{E} \setminus \{e\}, \phi_V, \phi_E), \\ \text{ModifyAttr}(v, \mathbf{a}'_v) &: \text{update attribute vector of } v \text{ to } \mathbf{a}'_v, \\ \text{AddContest}(e', e'', s_j) &: \text{add a contested hyperedge for the conflict pair } (e', e'') \text{ over } s_j. \end{aligned}$$

Each operator is recorded with a timestamp, the identity of the community member who initiated it, and the narrative vertex (oral testimony or annotation session record) that evidences it.

Remark 7.2. **RemoveEdge** is the only destructive operator. It is used only when a community member explicitly retracts a prior claim (as opposed to adding a competing claim). Community sessions are facilitated to distinguish these two cases carefully: retracting is different from contesting.

7.2 Active learning over the hypergraph

At each iteration of the validation loop, the annotation session should prioritise those structural instances whose reconstruction is most uncertain, so that community time is allocated to where it has the highest impact on the proposed design.

Definition 7.3 (Instance reconstruction uncertainty). For structural instance s_j , define the *reconstruction uncertainty* as

$$u_{s_j} = \frac{1}{|\mathcal{G}_{s_j}|} \sum_{k: \psi(k)=j} (\text{Var}[\boldsymbol{\mu}_k^{(t_2)}] + \text{Var}[\alpha_k^{(t_2)}]),$$

estimated by running M forward passes of the reverse diffusion process and computing empirical variances over the resulting Gaussian parameter samples.

The active learning criterion selects for annotation the instances $\mathcal{A}_{\text{query}} \subset \mathcal{S}$ satisfying at least one of:

1. $u_{s_j} > \tau_u$ (high geometric uncertainty), where τ_u is a threshold set at the 80th percentile of all instance uncertainties,
2. s_j is structurally incomplete (no ownership or testimony hyperedge incident), or
3. s_j is incident to an active contested hyperedge and the contestation has not yet been annotated in the current session.

Remark 7.4. The active learning loop is not intended to converge in one session. A typical annotation session covers a village or sub-village, takes 2–4 hours, and terminates when either the community is fatigued or the annotation budget is exhausted. Multiple sessions over multiple days, interleaved with re-generation runs, constitute the operational workflow.

7.3 The annotation interface

The annotation interface renders $\mathcal{G}^{(t_2)*}$ as a 3D interactive scene (via the differentiable rasteriser of Section 2.1) on a tablet device. Community members interact through the following modalities:

- **Tap to select:** selecting a structural instance highlights its governance hyperedges and renders the corresponding testimonies and actor associations in a side panel.
- **Tap to annotate:** attaching an oral note (audio, automatically transcribed) or a text label to the selected instance. The system creates a new narrative vertex $n \in \mathcal{N}$ and a testimony hyperedge linking it to the selected structure.
- **Dispute flag:** flagging a displayed ownership attribution as incorrect or contested, triggering AddContest.
- **Rule assertion:** asserting a governance rule over a set of selected structures, triggering AddNode for the rule vertex and AddEdge for the governance hyperedge.

8 On-Site Data Collection Protocol

This section describes the operational protocol for collecting the data that initialises both the spatial and semantic layers from on-site fieldwork in Morocco’s High Atlas. The protocol is designed for a single researcher with a tablet device, a consumer-grade drone, and Tachelhit-speaking local collaborators.

8.1 Drone survey design

The drone survey recovers the posed image collections required for $\mathcal{G}^{(t_1)}$ reconstruction (Section 3.2).

Flight protocol. For each village unit, execute three complementary flight patterns:

1. **Nadir grid:** a lawnmower pattern at 60 m above ground level with 80% forward and 70% side overlap. This provides uniform vertical coverage and is the primary input to COLMAP.
2. **Oblique orbit:** a circular orbit at 30 m AGL and 30° camera tilt, executed at 8 azimuthal positions. This recovers facade geometry that nadir imagery misses in dense earthen settlements.
3. **Targeted close-range:** manual flight at 5–15 m for partially collapsed or architecturally significant structures identified during the nadir survey. Images taken at multiple angles and distances.

Ground control points. A minimum of four identifiable ground control points (GCPs) per village, marked and geolocated by differential GPS, are required to anchor the reconstructed coordinate frame in a real-world geodetic datum. GCPs enable inter-village scenes to be registered to a common coordinate system, necessary when a single water channel connects structures across village boundaries.

8.2 Historical imagery recovery

Historical imagery for $\mathcal{G}^{(t_0)}$ is assembled from four sources in order of preference:

1. **Community member devices:** personal photographs and videos on mobile phones. A structured photo-elicitation session at the start of each community annotation session invites members to share relevant media. Photographs are geolocated if the device embedded GPS metadata, or approximately localised by matching to the t_1 reconstruction.
2. **Tourism and social media archives:** images of High Atlas villages are abundant in tourism photography platforms. A targeted crawl of geo-tagged images in the affected region, filtered to pre-September 2023, yields a substantial uncontrolled photo collection suitable for COLMAP.
3. **Institutional archives:** Getty Images, UNESCO World Heritage documentation, HERA-Morocco survey photographs, and the French colonial cartographic archive (Institut Géographique National) for villages with heritage designations.
4. **Satellite and aerial imagery:** Google Earth historical imagery (where available at sufficient resolution), Maxar WorldView-2/3, and SPOT archival scenes.

For uncontrolled photographs (sources 1 and 2), camera poses are unknown and must be recovered jointly by COLMAP. This is feasible when the image count exceeds approximately 30 per structure, coverage is multi-directional, and the scene has sufficient texture; earthen kasbahs with decorative facades generally satisfy these conditions.

8.3 Oral testimony pipeline

Oral testimonies are the primary source of narrative and rule vertices in $\mathcal{H}$. The pipeline has four stages.

1. **Elicitation:** structured interviews conducted in Tachelhit [تاشلحيت] by trained local collaborators. The interview guide follows spatial anchoring: the interviewee is shown the 3D render of the scene and asked to describe selected structures (“whose was this?”, “what was it used for?”, “what rules governed its use?”). Interviews are audio-recorded with consent.
2. **Transcription:** audio is transcribed into Tachelhit text, using a combination of human transcribers and multilingual ASR systems. We note that Tachelhit and other Tamazight varieties are low-resource languages: no robust dedicated ASR system currently exists for the dialects of the High Atlas. The practical workflow therefore relies on human transcription as the primary path, with ASR as a first-pass diarisation tool.
3. **Spatial grounding:** each transcribed statement is linked to one or more structural instances identified by the annotator during elicitation. A **statement** is added as a narrative vertex $n \in \mathcal{N}$, and testimony hyperedges connect it to the identified structural vertices.
4. **Rule extraction:** rule-relevant statements (those asserting permissions, prohibitions, or obligations with respect to spatial elements) are identified by the local collaborator and encoded as rule vertices $r \in \mathcal{R}$ with the structured format of Definition 4.4. This is a knowledge engineering task requiring community validation.

8.4 Structured annotation sessions

Community annotation sessions operationalise the interfaces described in Section 7.3. A session follows this structure:

1. **Orientation** (15 min): the facilitator introduces the 3D render of the current reconstruction proposal, explains what an annotation contributes, and establishes consent.
2. **Photo elicitation** (20 min): participants share historical photographs for the t_0 layer.
3. **Guided annotation** (90–120 min): active-learning query set $\mathcal{A}_{\text{query}}$ is presented one instance at a time. Participants tap, speak, dispute, and assert.
4. **Conflict documentation** (20 min): the facilitator explicitly elicits any disputed claims not surfaced during guided annotation, using a structured prompt (“are there any structures here whose ownership or use you would describe differently from what is shown?”).
5. **Closure** (10 min): hypergraph updates are summarised in plain language and confirmed by participants. The session produces a signed record of consented annotations.

9 Open Problems and Research Directions

The framework as specified in Sections 3 to 7 is mathematically complete in its formal structure, but several components require substantial further development before a deployable system can be built. We enumerate the most important open problems.

O1. Gaussian instance segmentation at architectural scale. The instance assignment map ψ (Definition 3.6) is formally defined, but computing it reliably for a dense village-scale 4DGS scene remains an open problem. Gaussian Grouping [20] demonstrates instance-level Gaussian segmentation for object-centric scenes; extending it to partially collapsed earthen architecture with irregular boundaries and severe occlusion requires domain-adapted training data that does not currently exist.

O2. Tachelhit NLP for rule extraction. Rule vertex construction from transcribed testimonies (Section 8.3, Stage 4) requires extracting structured deontic assertions from Tachelhit text. Multilingual pre-trained language models such as XLM-R [21] provide a weak prior for the Tamazight language family but have very limited coverage of Tachelhit. A dedicated annotated corpus of governance testimonies in Tachelhit, coupled with few-shot prompting of a multilingual language model, is likely the most tractable near-term approach.

O3. Constraint function design. The violation measure $c_r(\mathcal{G}', \mathcal{H})$ in (14) requires a rule-specific function f_r that evaluates whether a geometric configuration of Gaussian instance s_j satisfies or violates rule r . For spatial rules (“the water channel must have a continuous unobstructed path”), f_r can be defined geometrically. For social rules (“reconstruction of the granary loft must be led by the Ait Mansour lineage”), f_r cannot be evaluated from the Gaussian scene alone and requires actor-state information from the hypergraph. A systematic ontology of constraint function types and their computational implementations is needed.

O4. Cross-attention vs. concatenation in the joint embedding. The joint embedding (10) uses a simple concatenation of projected geometric and semantic features. A cross-attention mechanism in which the HGNN embeddings attend over the individual Gaussian positions (rather than their aggregated feature) could produce richer representations, particularly for large instances with complex internal spatial structure. Whether this justifies the additional computational cost is an empirical question.

O5. λ_1 , λ_2 calibration. The penalty weights λ_1 and λ_2 in (11) determine the trade-off between reconstruction fidelity, structural soundness, and governance compliance. A community-in-the-loop calibration procedure, in which designs at several (λ_1, λ_2) values are presented and ranked by community members, would replace the current ad hoc tuning. This connects to the multi-objective optimisation literature [22] and to participatory design methodology [23, 24].

O6. Convergence of the validation loop. The community validation loop of Section 7 is described operationally, but its convergence properties are uncharacterised. Does the sequence of hypergraphs $\mathcal{H}^{(0)}, \mathcal{H}^{(1)}, \dots$ generated by successive community sessions converge to a fixed point? Under what conditions on the update operators (Definition 7.1) and the community’s elicitation process? These questions connect to the theory of iterated belief revision and hypergraph dynamics.

O7. Multi-village scale. The framework is specified at the level of a single village or sub-village. The High Atlas reconstruction context involves many villages sharing governance structures (Agdal territories, valley-level water systems) that span village boundaries. Federated hypergraphs with cross-village governance edges, and corresponding multi-scene 4DGS models registered to a common geodetic frame, are the natural extension.

10 Conclusion

We have presented a layered mathematical framework for governance-aware generative landscape reconstruction in the context of Morocco’s High Atlas. The framework couples two primary representations: a 4D Gaussian splatting model that provides a geometrically faithful, multi-keyframe account of the physical landscape, and a typed governance hypergraph that encodes the social, cultural, and normative relationships constituting Amazigh land tenure and heritage governance.

The formal contributions, enumerated in Section 1.5, establish the precise mathematical structures required by each layer and their composition. Proposition 4.18 proves that conflict-preserving hyperedge representation is not a design preference but a mathematical requirement: collapsing contested claims expands the feasible reconstruction set, admitting designs that violate the governance rights of non-selected claimants. The constrained generation objective (11) corrects the independence assumption embedded in prior flat-sum formulations by routing governance losses through the co-functional hyperedge structure of $\mathcal{H}$.

Seven open problems (Section 9) mark the principal research frontiers: Gaussian instance segmentation at architectural scale, low-resource NLP for Tachelhit, constraint function design, embedding architecture, penalty calibration, validation-loop convergence, and multi-village federation. Progress on any of these would materially advance the system toward field deployment.

The work is motivated by and will be validated in the High Atlas communities affected by the 2023 Al-Haouz earthquake. Our commitment to conflict-preserving representation, to

community co-design of the hypergraph, and to the active learning annotation loop reflects a deeper conviction: the mathematical formalism must serve the community’s own governance, not substitute for it.

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